VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Big Data Analytics

Submitted by

Sharada Koundinya (1BM23CS310)

in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
Jan-2026 to May-2026

B. M. S. College of Engineering,

Bull Temple Road, Bangalore 560019

(Affiliated To Visvesvaraya Technological University, Belgaum)

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Lab work entitled "Big Data Analytics" carried out by **Sharada Koundinya (1BM23CS310),** who is a bona fide student of **B. M. S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a **Big Data Analytics - (23CS6PCBDA)** work prescribed for the said degree.

Prof. Laxman Singh
Assistant Professor
Department of CSE
BMSCE, Bengaluru

Dr. Kavitha Sooda
Professor and Head
Department of CSE
BMSCE, Bengaluru

INDEX SHEET

Sl. No.	Experiment Title	Page No.
1.	MongoDB CRUD Operations	1-5
2.	Cassandra Employee	6-7
3.	Cassandra Library	8-9
4.	Hadoop Commands	10-11
5.	Hadoop Word Count	12-14
6.	Hadoop Weather program	15-20
7.	Hadoop Top N Program	21-24
8.	Scala Program	25
9.	Spark program	26
10.	Open Ended question	27-28

Course Outcome

CO1	Apply the concepts of NoSQL, Hadoop, Spark for a given task
CO2	Analyze data analytic techniques for a given problem
CO3	Conduct experiments using data analytics mechanisms for a given problem

LAB 1
MongoDB- CRUD Demonstration

Student database

1. Create a database “Student” with the following attributes Rollno, Age, ContactNo, Email-Id.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.createCollection("Student");
{ ok: 1 }
```

2. Insert appropriate values (at least 5)

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:1,Age:21,Cont:9876,email:"antara.de9@gmail.com"});
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ec7c840f42b4a05530") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:2,Age:22,Cont:9976,email:"anushka.de9@gmail.com"});

{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05531") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:3,Age:21,Cont:5576,email:"anubhav.de9@gmail.com"});

{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05532") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:4,Age:20,Cont:4476,email:"pani.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05533") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:10,Age:23,Cont:2276,email:"rekha.de9@gmail.com"});
{
  acknowledged: true,
```

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:10,Age:23,Cont:2276,email:"rekha.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82f47c840f42b4a05534") }
}
```

3. View the data

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.find()
[
  {
    _id: ObjectId("660a82ec7c840f42b4a05530"),
    RollNo: 1,
    Age: 21,
    Cont: 9876,
    email: 'antara.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05531"),
    RollNo: 2,
    Age: 22,
    Cont: 9976,
    email: 'anushka.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05532"),
    RollNo: 3,
    Age: 21,
    Cont: 5576,
    email: 'anubhav.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05533"),
    RollNo: 4,
    Age: 20,
    Cont: 4476,
    email: 'pani.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82f47c840f42b4a05534"),
    RollNo: 10,
    Age: 23,
    Cont: 2276,
    email: 'rekha.de9@gmail.com'
  }
]
```

4. Write a query to update the Email-Id of a student with rollno 10.

```
  },
  {
    _id: ObjectId("660a83337c840f42b4a05535"),
    RollNo: 11,
    Age: 22,
    Name: 'ABC',
    Cont: 2276,
    email: 'rea.de9@gmail.com'
  }
]
```

5. Replace the student name from “ABC” to “FEM” of rollno 11.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.update({RollNo:11,Name:"ABC"},{$set:{Name:"FEM"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

6. Export the created table into local file system

```
C:\Users\BMSCECSE>mongoexport mongodb+srv://antararc:Test1234@cluster0.mfnfeys.mongodb.net/myDB  --collection=Student --out C:\Users\BMSCECSE\Downloads\output.json
2023-01-12T15:15:56.383+0530    connected to: mongodb+srv://[**REDACTED**]@cluster0.mfnfeys.mongodb.net/myDB
2023-01-12T15:15:56.497+0530    exported 7 records
```

7. Drop the table

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.drop();
true
```

8. Import a given csv dataset from local file system into mongodb collection.

```
C:\Users\BMSCECSE>mongoimport  mongodb+srv://antararc:Test1234@cluster0.mfnfeys.mongodb.net/myDB --collection=New_Student  --type json --file C:\Users\BMSCECSE\Downloads\output.json
2023-01-12T15:17:35.523+0530    connected to: mongodb+srv://[**REDACTED**]@cluster0.mfnfeys.mongodb.net/myDB
2023-01-12T15:17:35.640+0530    7 document(s) imported successfully. 0 document(s) failed to import.
```

Customer database

1. Create a collection by name Customers with the following attributes.
   Cust_id, Acc_Bal, Acc_Type

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.createCollection("Customers");
{ ok: 1 }
```

2. Insert at least 5 values into the table

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:1,Balance:200, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b47c840f42b4a05536") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:1,Balance:1000, Type:"Z"})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b47c840f42b4a05537") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:100, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b47c840f42b4a05538") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:1000, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b57c840f42b4a05539") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:500, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b57c840f42b4a0553a") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:50, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b57c840f42b4a0553b") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:3,Balance:500, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b77c840f42b4a0553c") }
}
```

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:50, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b57c840f42b4a0553b") }
}
```

3. Write a query to display those records whose total account balance is greater than 1200 of account type ‘Z’ for each customer_id.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.aggregate (
... {$match:{Type:"Z"}},
...
... {$group : { _id : "$cust_id",
...
... TotAccBal :{$sum:"$Balance"} } },
... {$match:{TotAccBal:{$gt:1200}}});
```

4. Determine Minimum and Maximum account balance for each customer_id.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.aggregate (
...
... {$group : { _id : "$cust_id",
...
... minAccBal :{$min:"$Balance"},
... maxAccBal :{$max:"$Balance"} }});
[
  { _id: 2, minAccBal: 50, maxAccBal: 1000 },
  { _id: 1, minAccBal: 200, maxAccBal: 1000 },
  { _id: 3, minAccBal: 500, maxAccBal: 500 }
]
```

5. Drop the table

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.drop()
true
```

LAB 2
Perform the following DB operations using Cassandra

Employee Database

1. Create a keyspace by name Employee

```
cqlsh:library> CREATE KEYSPACE Employee WITH REPLICATION = { 'class' : 'SimpleStrategy', 'replication_factor' : 1 };
cqlsh:library>
```

2. Create a column family by name Employee-Info with attributes Emp_Id Primary Key,

```
cqlsh:employee>
cqlsh:employee> CREATE TABLE Employee_Info (
           ...     Emp_Id int PRIMARY KEY,
           ...     Emp_Name text,
           ...     Designation text,
           ...     Date_of_Joining date,
           ...     Salary decimal,
           ...     Dept_Name text
           ... );
```

Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name

3. Insert the values into the table in batch

```
cqlsh:employee> BEGIN BATCH
           ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
           ... VALUES (101, 'John Doe', 'Manager', '2023-01-01', 50000, 'HR');
           ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
           ... VALUES (121, 'Jane Smith', 'Developer', '2023-02-01', 60000, 'IT');
           ... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info SET Emp_Name = 'Jane Johnson', Dept_Name = 'Engineering' WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name   | designation | emp_name     | salary
--------+-----------------+-------------+-------------+--------------+--------
    121 |      2023-02-01 | Engineering |   Developer | Jane Johnson |  60000
    101 |      2023-01-01 |          HR |     Manager |     John Doe |  50000

(2 rows)
```

4. Update Employee name and Department of Emp-Id 121

5. Sort the details of Employee records based on salary

```
cqlsh:employee> paging off
Disabled Query paging.
cqlsh:employee> SELECT * FROM Employee_Info WHERE Emp_Id IN (121,101) ORDER BY Salary ALLOW FILTERING;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectA', 'ProjectB'} WHERE Emp_Id = 101 and salary=50000;
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectC'} WHERE Emp_Id = 121 and salary=60000;
cqlsh:employee> select * from Employee_Info;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name   | projects
--------+--------+-----------------+-----------+-------------+------------+--------------------------
    121 |  60000 |      2023-02-01 |        IT |   Developer | Jane Smith |             {'ProjectC'}
    101 |  50000 |      2023-01-01 |        HR |     Manager |   John Doe | {'ProjectA', 'ProjectB'}

(2 rows)
```

6. Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
7. Update the altered table to add project names.

```
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectA', 'ProjectB'} WHERE Emp_Id = 101 and salary=50000;
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectC'} WHERE Emp_Id = 121 and salary=60000;
cqlsh:employee> select * from Employee_Info;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name   | projects
--------+--------+-----------------+-----------+-------------+------------+--------------------------
    121 |  60000 |      2023-02-01 |        IT |   Developer | Jane Smith |             {'ProjectC'}
    101 |  50000 |      2023-01-01 |        HR |     Manager |   John Doe | {'ProjectA', 'ProjectB'}

(2 rows)
```

8. Create a TTL of 15 seconds to display the values of Employees.

```
cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name) VALUES (102, 'Jane Smit
h', 'Developer', '2022-06-03', 60000, 'IT') USING TTL 15;
cqlsh:employee> select ttl(Emp_Name) from Employee_Info where Emp_id=102;

 ttl(emp_name)
---------------
             14

(1 rows)
```

LAB 3

Perform the following DB operations using Cassandra

1. Create a keyspace by name Library

```
cqlsh> CREATE KEYSPACE Library WITH REPLICATION = { 'class' : 'SimpleStrategy', 'replication_factor' : 1 };
cqlsh> show keyspaces;
Improper show command.
cqlsh> use Library;
cqlsh:library>
```

2. Create a column family by name Library-Info with attributes
Stud_Id Primary Key, Counter_value of type Counter,
Stud_Name, Book-Name, Book-Id, Date_of_issue

```
cqlsh:library> CREATE TABLE Library_Info (Stud_Id int PRIMARY KEY,Counter_value counter,Stud_Name text,Book_Name text,Book_Id text,Da
te_of_issue timestamp);
InvalidRequest: Error from server: code=2200 [Invalid query] message="Cannot mix counter and non counter columns in the same table"
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id int PRIMARY KEY,
           ...     Stud_Name text,
           ...     Book_Name text,
           ...     Book_Id text,
           ...     Date_of_issue timestamp
           ... );
cqlsh:library> CREATE TABLE Library_Counters (
           ...     Stud_Id int PRIMARY KEY,
           ...     Counter_value counter
           ... );
cqlsh:library>
```

3. Insert the values into the table in batch

```
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue) VALUES (112, 'John Doe', 'BDA', 'B001
', '2023-01-01');
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue) VALUES (113, 'Jane Smith', 'ML', 'B00
2', '2023-01-02');
           ... APPLY BATCH;
```

4. Display the details of the table created and increase the value of the counter

```
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue                   | stud_name
---------+---------+-----------+---------------------------------+------------
     113 |    B002 |        ML | 2023-01-02 00:00:00.000000+0000 | Jane Smith
     112 |    B001 |       BDA | 2023-01-01 00:00:00.000000+0000 |   John Doe

(2 rows)
cqlsh:library> SELECT * FROM Library_Counters;

 stud_id | counter_value
---------+---------------
     113 |             1
     112 |             1

(2 rows)
```

5. Write a query to show that a student with id 112 has taken a book “BDA” 2 times.

```
cqlsh:library> UPDATE Library_Counters SET Counter_value = Counter_value + 1 WHERE Stud_Id = 112;
cqlsh:library> SELECT * FROM Library_Counters WHERE Stud_Id = 112;

 stud_id | counter_value
---------+---------------
     112 |             2

(1 rows)
```

6. Export the created column to a csv file

```
cqlsh:library> COPY Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue) TO 'file.csv' WITH HEADER = TRUE;
Using 11 child processes

Starting copy of library.library_info with columns [stud_id, stud_name, book_name, book_id, date_of_issue].
Processed: 2 rows; Rate:      10 rows/s; Avg. rate:       6 rows/s
2 rows exported to 1 files in 0.374 seconds.
cqlsh:library> COPY Library_Counters (Stud_Id, Counter_value) FROM 'library_counters.csv' WITH HEADER = TRUE;
Using 11 child processes
```

7. Import a given csv dataset from local file system into Cassandra column family

```
cqlsh:library> copy library_info(Stud_Id,Stud_Name,Book_Name,Book_Id,Date_of_issue) from 'file.csv' with header=true;
Using 7 child processes

Starting copy of library.library_info with columns [stud_id, stud_name, book_name, book_id, date_of_issue].
Processed: 2 rows; Rate:       2 rows/s; Avg. rate:       4 rows/s
2 rows imported from 1 files in 0.513 seconds (0 skipped).
cqlsh:library> select * from library_info;

 stud_id | book_id | book_name | date_of_issue                   | stud_name
---------+---------+-----------+---------------------------------+------------
     113 |    B002 |        ML | 2023-01-02 00:00:00.000000+0000 | Jane Smith
     112 |    B001 |       BDA | 2023-01-01 00:00:00.000000+0000 |   John Doe
```

LAB 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

1. mkdir
2. ls

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -mkdir /bda_hadoop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /
Found 1 items
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 14:37 /bda_hadoop
```

3. put

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -put /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /bda_hadoop
Found 1 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /bda_hadoop/file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file.txt
Hello!!!
```

4. copyFromLocal

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello!!!
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

5. get

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -get /bda_hadoop/file.txt /home/hadoop/Desktop/downloaded_file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -getmerge /bda_hadoop/file.txt /bda_hadoop/file_cp_local.txt /home/hadoop/Desktop/downloaded_file.txt
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -getfacl /bda_hadoop/
# file: /bda_hadoop
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
```

6. copyToLocal

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyToLocal /bda_hadoop/file.txt /home/hadoop/Desktop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /bda_hadoop /abc
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /abc
Found 2 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /abc/file.txt
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /hello/ /hadoop_lab
cp: `/hello/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

7. cat

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello!!!
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

8. mv

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /bda_hadoop /abc
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /abc
Found 2 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /abc/file.txt
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
```

9. cp

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /hello/ /hadoop_lab
cp: `/hello/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

LAB 5
Implement WordCount Program on Hadoop framework

Word Count Program:

Mapper code: You have to copy paste this program into the WCMapper Java Class file

```
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;
public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text,
Text, IntWritable> {
    public void map(LongWritable key, Text value, OutputCollector<Text, IntWritable> output,
Reporter rep) throws IOException {
        String line = value.toString();
        for (String word : line.split(" ")) {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}
```

Reducer Code: You have to copy paste this program into the WCReducer Java Class file

```
import java.io.IOException;
import java.util.Iterator;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import  org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;
public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text,
IntWritable> {
    public void reduce(Text key, Iterator<IntWritable> value, OutputCollector<Text, IntWritable>
output, Reporter rep) throws IOException {
        int count = 0;
        while (value.hasNext()) {
            IntWritable i = value.next();
            count += i.get();
```

```
        }
        output.collect(key, new IntWritable(count));
    }
}
```

Driver Code: You have to copy paste this program into the WCDriver Java Class file.

```
import java.io.IOException;
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;
public class WCDriver extends Configured implements Tool {
    public int run(String[] args) throws IOException {
        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }
        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("Word Count");
        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));
        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);
        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);
        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);
        JobClient.runJob(conf);
        return 0;
    }
    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println(exitCode);
    }
}
```

OUTPUT

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /input_dir/input_file.txt
Hello World
Hello Hadoop
This is Hadoop test file
C:\hadoop-3.3.0\sbin>hadoop jar C:\MapReduceClient.jar wordcount /input_dir /output_dir
2021-04-24 15:24:57,242 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2021-04-24 15:24:57,714 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging
/job_1619256355508_0002
2021-04-24 15:24:58,387 INFO input.FileInputFormat: Total input files to process : 1
2021-04-24 15:24:58,809 INFO mapreduce.JobSubmitter: number of splits:1
2021-04-24 15:24:59,255 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1619256355508_0002
2021-04-24 15:24:59,255 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-04-24 15:24:59,450 INFO conf.Configuration: resource-types.xml not found
2021-04-24 15:24:59,451 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-04-24 15:24:59,533 INFO impl.YarnClientImpl: Submitted application application_1619256355508_0002
2021-04-24 15:24:59,581 INFO mapreduce.Job: The url to track the job: http://LAPTOP-JG329ESD:8088/proxy/application_1619256355508_0002/
2021-04-24 15:24:59,582 INFO mapreduce.Job: Running job: job_1619256355508_0002
2021-04-24 15:25:12,857 INFO mapreduce.Job: Job job_1619256355508_0002 running in uber mode : false
2021-04-24 15:25:12,861 INFO mapreduce.Job:  map 0% reduce 0%
2021-04-24 15:25:19,985 INFO mapreduce.Job:  map 100% reduce 0%
2021-04-24 15:25:26,077 INFO mapreduce.Job:  map 100% reduce 100%
2021-04-24 15:25:32,181 INFO mapreduce.Job: Job job_1619256355508_0002 completed successfully
2021-04-24 15:25:32,284 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=85
                FILE: Number of bytes written=530945
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=162
                HDFS: Number of bytes written=51
```

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /output_dir/*
Hadoop  2
Hello   2
This    1
World   1
file    1
is      1
test    1

C:\hadoop-3.3.0\sbin>
```

<u>LAB 6</u>

From the following link extract the weather data
https://github.com/tomwhite/hadoop-Book/tree/master/input/ncdc/all

Create a Map Reduce program to
a) find average temperature for each year from the NCDC data set.

AverageDriver

```
package temp;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class AverageDriver {
   public static void main(String[] args) throws Exception {
      if (args.length != 2) {
         System.err.println("Please Enter the input and output parameters");
         System.exit(-1);
      }
      Job job = Job.getInstance();
      job.setJarByClass(AverageDriver.class);
      job.setJobName("Max temperature");
      FileInputFormat.addInputPath(job, new Path(args[0]));
      FileOutputFormat.setOutputPath(job, new Path(args[1]));
      job.setMapperClass(AverageMapper.class);
      job.setReducerClass(AverageReducer.class);
      job.setOutputKeyClass(Text.class);
      job.setOutputValueClass(IntWritable.class);
      System.exit(job.waitForCompletion(true) ? 0 : 1);
   }
}
```

AverageMapper

```
package temp;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
```

```
   public static final int MISSING = 9999;
      public void map(LongWritable key, Text value, Context context) throws IOException,
InterruptedException {
      int temperature;
      String line = value.toString();
      if (line.length() < 93) {
                return;
      }
      String year = line.substring(15, 19);
      if (line.charAt(87) == '+') {
         temperature = Integer.parseInt(line.substring(88, 92));
      } else {
         temperature = Integer.parseInt(line.substring(87, 92));
      }
      String quality = line.substring(92, 93);
      if (temperature != MISSING && quality.matches("[01459]")) {
         context.write(new Text(year), new IntWritable(temperature));
      }
   }
}
```

AverageReducer

```
package temp;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
   public void reduce(Text key, Iterable<IntWritable> values,
                  Context context) throws IOException, InterruptedException {
      int sum = 0;
      int count = 0;
      for (IntWritable value : values) {
         sum += value.get();
         count++;
      }
      if (count > 0) {
         int average = sum / count;
         context.write(key, new IntWritable(average));
      }
   }
}
```

OUTPUT

```
:\hadoop-3.3.0\sbin>hadoop jar C:\avgtemp.jar temp.AverageDriver /input_dir/temp.txt /avgtemp_outputdir
021-05-15 14:52:50,635 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
021-05-15 14:52:51,005 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
021-05-15 14:52:51,111 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging/job_1621060230696_0005
021-05-15 14:52:51,735 INFO input.FileInputFormat: Total input files to process : 1
021-05-15 14:52:52,751 INFO mapreduce.JobSubmitter: number of splits:1
021-05-15 14:52:53,073 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1621060230696_0005
021-05-15 14:52:53,073 INFO mapreduce.JobSubmitter: Executing with tokens: []
021-05-15 14:52:53,237 INFO conf.Configuration: resource-types.xml not found
021-05-15 14:52:53,238 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
021-05-15 14:52:53,312 INFO impl.YarnClientImpl: Submitted application application_1621060230696_0005
021-05-15 14:52:53,352 INFO mapreduce.Job: The url to track the job: http://LAPTOP-3G329ESD:8088/proxy/application_1621060230696_0005/
021-05-15 14:52:53,353 INFO mapreduce.Job: Running job: job_1621060230696_0005
021-05-15 14:53:06,640 INFO mapreduce.Job: Job job_1621060230696_0005 running in uber mode : false
021-05-15 14:53:06,643 INFO mapreduce.Job:  map 0% reduce 0%
021-05-15 14:53:12,758 INFO mapreduce.Job:  map 100% reduce 0%
021-05-15 14:53:19,860 INFO mapreduce.Job:  map 100% reduce 100%
021-05-15 14:53:25,967 INFO mapreduce.Job: Job job_1621060230696_0005 completed successfully
021-05-15 14:53:26,096 INFO mapreduce.Job: Counters: 54
      File System Counters
            FILE: Number of bytes read=72210
            FILE: Number of bytes written=674341
            FILE: Number of read operations=0
            FILE: Number of large read operations=0
            FILE: Number of write operations=0
            HDFS: Number of bytes read=894860
            HDFS: Number of bytes written=8
            HDFS: Number of read operations=8
            HDFS: Number of large read operations=0
            HDFS: Number of write operations=2
            HDFS: Number of bytes read erasure-coded=0
      Job Counters
            Launched map tasks=1
            Launched reduce tasks=1
            Data-local map tasks=1
            Total time spent by all maps in occupied slots (ms)=3782
```

```
C:\hadoop-3.3.0\sbin>hdfs dfs -ls /avgtemp_outputdir
Found 2 items
-rw-r--r--   1 Anusree supergroup          0 2021-05-15 14:53 /avgtemp_outputdir/_SUCCESS
-rw-r--r--   1 Anusree supergroup          8 2021-05-15 14:53 /avgtemp_outputdir/part-r-00000

C:\hadoop-3.3.0\sbin>hdfs dfs -cat /avgtemp_outputdir/part-r-00000
1901    46

C:\hadoop-3.3.0\sbin>
```

b) find the mean max temperature for every month

MeanMaxDriver.class

```
package meanmax;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class MeanMaxDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
```

```
        System.err.println("Please enter the input and output parameters");
        System.exit(-1);
    }
    Job job = Job.getInstance();
    job.setJarByClass(MeanMaxDriver.class);
    job.setJobName("Mean and Max Temperature");
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    job.setMapperClass(MeanMaxMapper.class);
    job.setReducerClass(MeanMaxReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
}
```

MeanMaxMapper.class

```
package meanmax;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> {
  public static final int MISSING = 9999;
  public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
    int temperature;
    String line = value.toString();
    if (line.length() < 93) return;
    String month = line.substring(19, 21);       if (line.charAt(87) == '+') {
        temperature = Integer.parseInt(line.substring(88, 92));
    } else {
        temperature = Integer.parseInt(line.substring(87, 92));
    }
    String quality = line.substring(92, 93);
    if (temperature != MISSING && quality.matches("[01459]")) {
        context.write(new Text(month), new IntWritable(temperature));
    }
  }
}
```

MeanMaxReducer.class

```
package meanmax;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
            throws IOException, InterruptedException {
        int maxTemp = 0;
        int totalTemp = 0;
        int count = 0;
        int days = 0;
        for (IntWritable value : values) {
            int temp = value.get();
            if (temp > maxTemp) {
                maxTemp = temp;
            }
            count++;
            if (count == 3) {
                totalTemp += maxTemp;
                maxTemp = 0;
                count = 0;
                days++;
            }
        }
        if (days > 0) {
            context.write(key, new IntWritable(totalTemp / days));
        } else {
            context.write(key, new IntWritable(0));
        }
    }
}
```

OUTPUT

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /meanmax_output/*
01      4
02      0
03      7
04      44
05      100
06      168
07      219
08      198
09      141
10      100
11      19
12      3

C:\hadoop-3.3.0\sbin>
```

LAB 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words

TopNDriver.class

```
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.util.GenericOptionsParser;
public class TopN {
   public static void main(String[] args) throws Exception {
      Configuration conf = new Configuration();
      String[] otherArgs = (new GenericOptionsParser(conf, args)).getRemainingArgs();
      if (otherArgs.length != 2) {
         System.err.println("Usage: TopN <in> <out>");
         System.exit(2);
      }
      Job job = Job.getInstance(conf);
      job.setJobName("Top N");
      job.setJarByClass(TopN.class);
      job.setMapperClass(TopNMapper.class);
      job.setReducerClass(TopNReducer.class);
      job.setOutputKeyClass(Text.class);
      job.setOutputValueClass(IntWritable.class);
      FileInputFormat.addInputPath(job, new Path(otherArgs[0]));
      FileOutputFormat.setOutputPath(job, new Path(otherArgs[1]));
      System.exit(job.waitForCompletion(true) ? 0 : 1);
   }
}
```

TopNCombiner.class

```
package samples.topn;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
```

```
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class TopNCombiner extends Reducer<Text, IntWritable, Text, IntWritable> {
   public void reduce(Text key, Iterable<IntWritable> values, Context context)
         throws IOException, InterruptedException {
      int sum = 0;
      for (IntWritable val : values)
         sum += val.get();
      context.write(key, new IntWritable(sum));
   }
}
```

TopNReducer.class

```
package samples.topn;
import java.io.IOException;
import java.util.HashMap;
import java.util.Map;
import java.util.LinkedHashMap;
import java.util.stream.Collectors;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class TopNReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
   private Map<Text, Integer> countMap = new HashMap<>();
      public void reduce(Text key, Iterable<IntWritable> values, Context context) throws
IOException, InterruptedException {
      int sum = 0;
      for (IntWritable val : values) {
         sum += val.get();
      }
            countMap.put(new Text(key), sum);
   }
   protected void cleanup(Context context) throws IOException, InterruptedException {
         Map<Text, Integer> sortedMap = countMap.entrySet().stream().sorted(Map.Entry.<Text,
Integer>comparingByValue().reversed()).limit(20).collect(Collectors.toMap(
            Map.Entry::getKey,
            Map.Entry::getValue,
            (e1, e2) -> e1,
            LinkedHashMap::new));
      for (Map.Entry<Text, Integer> entry : sortedMap.entrySet()) {
         context.write(entry.getKey(), new IntWritable(entry.getValue()));
      }
   }
```

```
}
```

TopNMapper.class

```
package samples.topn;
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
public class TopNMapper extends Mapper<Object, Text, Text, IntWritable> {
   private static final IntWritable one = new IntWritable(1);
   private Text word = new Text();
   private String tokens = "[_|$#<>\\^=\\[\\]\\*/\\\\,;,.\\-:()?!\"']";
     public void map(Object key, Text value, Mapper<Object, Text, Text, IntWritable>.Context
context)
         throws IOException, InterruptedException {
      String cleanLine = value.toString().toLowerCase().replaceAll(this.tokens, " ");
      StringTokenizer itr = new StringTokenizer(cleanLine);
      while (itr.hasMoreTokens()) {
         this.word.set(itr.nextToken().trim());
         context.write(this.word, one);
      }
   }
}
```

OUTPUT

```
C:\hadoop-3.3.0\sbin>jps
11072 DataNode
20528 Jps
5620 ResourceManager
15532 NodeManager
6140 NameNode

C:\hadoop-3.3.0\sbin>hdfs dfs -mkdir /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - Anusree supergroup          0 2021-05-08 19:46 /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -copyFromLocal C:\input.txt /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /input_dir
Found 1 items
-rw-r--r--   1 Anusree supergroup         36 2021-05-08 19:48 /input_dir/input.txt

C:\hadoop-3.3.0\sbin>hdfs dfs -cat /input_dir/input.txt
hello
world
hello
hadoop
bye
```

```
C:\hadoop-3.3.0\sbin>hadoop jar C:\sort.jar samples.topn.TopN /input_dir/input.txt /output_dir
2021-05-08 19:54:54,582 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2021-05-08 19:54:55,291 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging/job_1620483374279_0001
2021-05-08 19:54:55,821 INFO input.FileInputFormat: Total input files to process : 1
2021-05-08 19:54:56,261 INFO mapreduce.JobSubmitter: number of splits:1
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1620483374279_0001
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-05-08 19:54:56,843 INFO conf.Configuration: resource-types.xml not found
2021-05-08 19:54:56,843 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-05-08 19:54:57,387 INFO impl.YarnClientImpl: Submitted application application_1620483374279_0001
2021-05-08 19:54:57,587 INFO mapreduce.Job: The url to track the job: http://LAPTOP-JG329ESD:8088/proxy/application_1620483374279_0001/
2021-05-08 19:54:57,508 INFO mapreduce.Job: Running job: job_1620483374279_0001
2021-05-08 19:55:13,792 INFO mapreduce.Job: Job job_1620483374279_0001 running in uber mode : false
2021-05-08 19:55:13,794 INFO mapreduce.Job:  map 0% reduce 0%
2021-05-08 19:55:20,020 INFO mapreduce.Job:  map 100% reduce 0%
2021-05-08 19:55:27,116 INFO mapreduce.Job:  map 100% reduce 100%
2021-05-08 19:55:33,199 INFO mapreduce.Job: Job job_1620483374279_0001 completed successfully
2021-05-08 19:55:33,334 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=65
                FILE: Number of bytes written=530397
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=142
                HDFS: Number of bytes written=31
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
```

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /output_dir/*
hello   2
hadoop  1
world   1
bye     1

C:\hadoop-3.3.0\sbin>
```

LAB 8&9

1. Write a Scala program to print numbers from 1 to 100 using for loop.

```
object ExampleForLoop1 {
   def main(args: Array[String]) {
      for(counter <- 1 to 100) {
         print(counter + " ")
      }
      println()
   }
}
```

OUTPUT

```
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ nano ExampleForLoop1.scala
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ scala -version
Scala code runner version 2.11.12 -- Copyright 2002-2017, LAMP/EPFL
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ nano ExampleForLoop1.scala
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ scalac ExampleForLoop1.scala
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ scala ExampleForLoop1
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71
72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

2. Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

```
from pyspark import SparkContext
sc = SparkContext("local", "SimpleWordCount")

# Read file
rdd = sc.textFile("file.txt")  # Replace with your filename

# Count words
counts = (rdd.flatMap(lambda line: line.split())
          .map(lambda word: (word, 1))
          .reduceByKey(lambda a, b: a + b)
          .filter(lambda x: x[1] > 4))

# Show result
for word, count in counts.collect():
   print(word, count)

sc.stop()
```

OUTPUT

```
scala> val data = sc.textFile("/home/bmscecse/Desktop/sparkdata.txt")
data: org.apache.spark.rdd.RDD[String] = /home/bmscecse/Desktop/sparkdata.txt MapPartitionsRDD[9] at textFile at <console>:24

scala> val data = sc.textFile("/home/bmscecse/Desktop/sparkdata.txt")
data: org.apache.spark.rdd.RDD[String] = /home/bmscecse/Desktop/sparkdata.txt MapPartitionsRDD[11] at textFile at <console>:24

scala> val mapdata = splitdata.map(word => (word, 1))
mapdata: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[12] at map at <console>:25

scala> val reducedata = mapdata.reduceByKey(_ + _)
reducedata: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[13] at reduceByKey at <console>:25

scala> val filteredWords = reducedata.filter{ case (word, count) => count > 4 }
filteredWords: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[14] at filter at <console>:25

scala> filteredWords.collect.foreach(println)

scala> val data = sc.textFile("/home/bmscecse/Desktop/sparkdata.txt")
data: org.apache.spark.rdd.RDD[String] = /home/bmscecse/Desktop/sparkdata.txt MapPartitionsRDD[16] at textFile at <console>:24

scala> val splitdata = data.flatMap(line => line.split(" "))
splitdata: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[17] at flatMap at <console>:25

scala> val mapdata = splitdata.map(word => (word, 1))
mapdata: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[18] at map at <console>:25

scala> val reducedata = mapdata.reduceByKey(_ + _)
reducedata: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[19] at reduceByKey at <console>:25

scala> val filteredWords = reducedata.filter{ case (word, count) => count >= 4 }
filteredWords: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[20] at filter at <console>:25

scala> filteredWords.collect.foreach(println)
(hello,4)

scala> filteredWords.saveAsTextFile("/home/bmscecse/Desktop/filtered_words")

scala> val sortedWords = reducedata.sortBy(_._2, ascending = false)
sortedWords: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[26] at sortBy at <console>:25

scala> sortedWords.collect.foreach(println)
(hello,4)
(spark,3)
(world,2)
(big,1)
(data,1)
(,1)

scala> val countWords = filteredWords.count()
countWords: Long = 1

scala> println(s"Number of words with count >= 4: $countWords")
Number of words with count >= 4: 1

scala>
```

LAB 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

```
import org.apache.spark.SparkConf
import org.apache.spark.streaming.{Seconds, StreamingContext}
import org.apache.spark.streaming.StreamingContext._

object SimpleStreamingCleaner {
  def main(args: Array[String]): Unit = {
    // Create Spark configuration and StreamingContext
    val conf = new SparkConf().setMaster("local[2]").setAppName("StreamingTextCleaner")
    val ssc = new StreamingContext(conf, Seconds(5))

    // Connect to socket stream (e.g., using: `nc -lk 9999`)
    val lines = ssc.socketTextStream("localhost", 9999)

    // Define stop words
    val stopWords = Set("a", "an", "the", "is", "in", "on", "and", "of", "to", "this", "that")

    // Define basic lemmatization map (for demo only)
      val lemmaMap = Map("running" -> "run", "tests" -> "test", "files" -> "file", "words" ->
"word")

    // Clean the text stream
    val cleaned = lines.map(_.toLowerCase.trim) // lower case and trim
      .flatMap(_.split("\\s+"))                 // split into words
      .filter(word => !stopWords.contains(word)) // remove stop words
      .map(word => lemmaMap.getOrElse(word, word)) // basic lemmatization

    // Print the cleaned text
    cleaned.print()

    // Start the stream
    ssc.start()
    ssc.awaitTermination()
  }
}
```

OUTPUT

```
scala> import java.nio.file.{Files, Paths}
import java.nio.file.{Files, Paths}

scala>

scala> val path = "/home/bmscecse/Desktop/WC.txt"
path: String = /home/bmscecse/Desktop/WC.txt

scala> val exists = Files.exists(Paths.get(path))
exists: Boolean = true

scala> println(s"File exists? $exists")
File exists? true

scala> val textFile = sc.textFile("file:///home/bmscecse/Desktop/WC.txt")
textFile: org.apache.spark.rdd.RDD[String] = file:///home/bmscecse/Desktop/WC.txt MapPartitionsRDD[40] at textFile at <console>:26

scala>

scala> val words = textFile.flatMap(line => line.split("\\s+"))
words: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[41] at flatMap at <console>:27

scala>

scala> val pairs = words.map(word => (word, 1))
pairs: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[42] at map at <console>:27

scala>

scala> val counts = pairs.reduceByKey(_ + _)
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[43] at reduceByKey at <console>:27

scala>

scala> counts.collect().foreach{ case (word, count) => println(s"$word: $count") }
this: 3
is: 2
some: 1
: 1
hello: 5
testing: 1
just: 1
file: 3
test: 2
world: 4
contains: 1
a: 1
for: 1
words: 1

scala>
```